DAILY PRACTICE CALENDAR

Grade 5 Competition Math • 2027–2028

34 weeks • Monday–Friday • 10 problems a day (about 30–40 minutes)

Experience first. Symbols second. Understanding always.

How to Use This Calendar

- **Mon–Thu:** 10 problems on the day's skill: **8 new + 2 review** from earlier weeks. Start with the **Try** problem as the warm-up.
- **Friday:** Weekly Quiz of 10 mixed problems (8 this week + 2 review). Copy every missed problem into the **error log** and redo it next Monday.
- **Holidays and breaks** are marked. A holiday's skill moves to the weekend; break weeks are not counted in the 34 weeks.
- Answers to every **Try** problem are on the last page.

Year at a Glance

Wk	Dates	Topic
1	Aug 16–Aug 20	Problem-Solving Toolkit
2	Aug 23–Aug 27	Place Value & Digit Puzzles
3	Aug 30–Sep 3	Clever Calculation
4	Sep 6–Sep 10	Operations & Number Puzzles
5	Sep 13–Sep 17	Divisibility Rules
6	Sep 20–Sep 24	Factors, Primes & Prime Factorization
7	Sep 27–Oct 1	GCF & LCM
8	Oct 4–Oct 8	Counting Factors & Square Numbers
9	Oct 11–Oct 15	Remainders & Cycles
10	Oct 18–Oct 22	Review & Mini-Contest 1
11	Oct 25–Oct 29	Fraction Sense
12	Nov 1–Nov 5	Fraction Word Problems
13	Nov 8–Nov 12	Decimals & Percents
14	Nov 15–Nov 19	Ratio & Rate
	Nov 22–Nov 26	*Thanksgiving Break*
15	Nov 29–Dec 3	Averages
16	Dec 6–Dec 10	Number Patterns & Sequences
17	Dec 13–Dec 17	Growing Shape Patterns
	Dec 20–Dec 24	*Winter Break*
	Dec 27–Dec 31	*Winter Break*
18	Jan 3–Jan 7	Bar-Model Word Problems
19	Jan 10–Jan 14	Assumption & Work Backward
20	Jan 17–Jan 21	Mid-Year Mock Contest & Review
21	Jan 24–Jan 28	Angles & Polygons
22	Jan 31–Feb 4	Perimeter & Area
23	Feb 7–Feb 11	Area by Cut & Rearrange
24	Feb 14–Feb 18	Counting Figures
25	Feb 21–Feb 25	Cubes, Volume & Surface Area
26	Feb 28–Mar 3	Nets & Views
27	Mar 6–Mar 10	Systematic Counting
28	Mar 13–Mar 17	Multiplication Principle
29	Mar 20–Mar 24	Venn Diagrams
	Mar 27–Mar 31	*Spring Break*
30	Apr 3–Apr 7	Probability
31	Apr 10–Apr 14	Logic & Pigeonhole
32	Apr 17–Apr 21	Contest Strategy & AMC 8 Preview
33	Apr 24–Apr 28	Full Mock Contest & Review
34	May 1–May 5	Final Contest & Celebration

Units 1 & 6 · Unit 2 · Unit 3 · Unit 4 · Unit 5 · Wrap-Up · Break

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WEEK 1 • Aug 16–Aug 20, 2027

Unit 1: Problem-Solving & Number Sense

Problem-Solving Toolkit

Core: Read carefully, then pick a strategy: picture, table, work backward, guess-check-improve.

Mon Aug 16

Read & underline the question

TRY A book costs $7 more than a pen. Together they cost $19. How much is the book?

Tue Aug 17

Draw a picture

TRY 8 trees stand in a row, 5 m apart. How far is it from the first tree to the last?

Wed Aug 18

Make an organized list

TRY How many ways can you make 25¢ using only nickels and dimes?

Thu Aug 19

Work backward

TRY I double a number, then add 7, and get 31. What was my number?

Fri Aug 20

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 2 • Aug 23–Aug 27, 2027

Unit 1: Problem-Solving & Number Sense

Place Value & Digit Puzzles

Core: Use place value to build, compare and repair numbers.

Mon Aug 23

Place value to millions

TRY What is the value of the digit 7 in 3,472,605?

Tue Aug 24

Largest / smallest numbers

TRY Use 3, 8, 0, 5 once each. What is the largest even 4-digit number?

Wed Aug 25

Digit sums

TRY How many 2-digit numbers have a digit sum of 9?

Thu Aug 26

Missing-digit arithmetic

TRY Fill in the boxes: 4□7 + 25□ = 703.

Fri Aug 27

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 3 • Aug 30–Sep 3, 2027

Unit 1: Problem-Solving & Number Sense

Clever Calculation

Core: Look for friendly numbers before computing.

Mon Aug 30

Regrouping & compensation

TRY Compute 298 + 467 in your head.

Tue Aug 31

Friendly products

TRY Compute $25 \times 37 \times 4$.

Wed Sep 1

Distributive shortcuts

TRY Compute 99×46.

Thu Sep 2

Gauss pairing

TRY Find $1 + 2 + 3 + \cdots + 50$.

Fri Sep 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 4 • Sep 6–Sep 10, 2027

Unit 1: Problem-Solving & Number Sense

Operations & Number Puzzles

Core: Order of operations and puzzle-style reasoning with numbers.

Mon Sep 6

HOLIDAY: Labor Day

No class. Do this skill's 10 problems on the weekend:

Order of operations

TRY Compute $36 \div (4 + 5) \times 3 - 2$.

Tue Sep 7

Insert the operations

TRY Use $+, -, \times, \div$ and ( ) to make 4 4 4 4 = 7.

Wed Sep 8

Magic squares

TRY Place 1–9 in a 3×3 magic square. Which number must go in the center?

Thu Sep 9

Made-up operations

TRY $\star$ means "double the first number, then add the second." Find $5 \star 3$.

Fri Sep 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 5 • Sep 13–Sep 17, 2027 — Unit 2: Number Theory

Divisibility Rules — **Core:** Test divisibility quickly without dividing.

Mon Sep 13	Tue Sep 14	Wed Sep 15	Thu Sep 16	Fri Sep 17
Rules for 2, 4, 5, 8, 10	**Rules for 3 and 9**	**Rules for 6 and 11**	**Find the missing digit**	**Weekly Quiz**
TRY Is 7,536 divisible by 4? By 8?	TRY Is 48,213 divisible by 9?	TRY Is 5,071 divisible by 11?	TRY What digit fills 36$\square$4 so it is divisible by 9?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 6 • Sep 20–Sep 24, 2027 — Unit 2: Number Theory

Factors, Primes & Prime Factorization — **Core:** Break numbers into their prime building blocks.

Mon Sep 20	Tue Sep 21	Wed Sep 22	Thu Sep 23	Fri Sep 24
Factor pairs	**Primes to 100**	**Factor trees**	**Numbers with given properties**	**Weekly Quiz**
TRY List all factors of 36. How many are there?	TRY How many prime numbers are between 40 and 60?	TRY Write 360 as a product of primes.	TRY Smallest 3-digit number whose only prime factors are 2 and 3 (both used)?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 7 • Sep 27–Oct 1, 2027 — Unit 2: Number Theory

GCF & LCM — **Core:** Greatest common factor shares; least common multiple meets again.

Mon Sep 27	Tue Sep 28	Wed Sep 29	Thu Sep 30	Fri Oct 1
GCF	**LCM**	**Ladder method; GCF $\times$ LCM**	**GCF/LCM word problems**	**Weekly Quiz**
TRY Find the GCF of 48 and 72.	TRY Find the LCM of 12 and 18.	TRY GCF(24, 36) = 12. Use GCF $\times$ LCM = 24 $\times$ 36 to find the LCM.	TRY Bell A rings every 6 min, Bell B every 8 min. Both ring at 9:00. When together next?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 8 • Oct 4–Oct 8, 2027 — Unit 2: Number Theory

Counting Factors & Square Numbers — **Core:** Count factors from the prime factorization.

Mon Oct 4	Tue Oct 5	Wed Oct 6	Thu Oct 7	Fri Oct 8
Counting factors	**Perfect squares**	**Exactly 2, 3, 4 factors**	**Lockers problem**	**Weekly Quiz**
TRY How many factors does 72 have?	TRY How many perfect squares are from 1 to 200?	TRY Which numbers less than 50 have exactly 3 factors?	TRY 100 closed lockers; student k flips every kth locker (k = 1 to 100). How many end open?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 9 • Oct 11–Oct 15, 2027 — Unit 2: Number Theory

Remainders & Cycles **Core:** Repeating patterns are solved with remainders.

Mon Oct 11
Remainders
TRY What is the remainder when 2027 is divided by 7?

Tue Oct 12
Repeating patterns
TRY ABCDABCD… What is the 100th letter?

Wed Oct 13
Day-of-week problems
TRY Today is Monday. What day will it be in 100 days?

Thu Oct 14
Units digits of powers
TRY What is the units digit of 3^{50}?

Fri Oct 15
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 10 • Oct 18–Oct 22, 2027 — Unit 2: Number Theory

Review & Mini-Contest 1 **Core:** Pull Units 1–2 together and practice contest habits.

Mon Oct 18
Review: strategies & clever calculation
TRY Compute 125×32 cleverly.

Tue Oct 19
Review: divisibility & factors
TRY Find every digit for $7\square2$ so it is divisible by both 3 and 4.

Wed Oct 20
Review: GCF/LCM & cycles
TRY What is the units digit of 7^{23}?

Thu Oct 21
Timed set: 10 problems in 20 min
TRY Find the LCM of 8, 12 and 15.

Fri Oct 22
Mini-Contest 1
Units 1–2, timed. Afterward, start an error log of every missed problem.

WEEK 11 • Oct 25–Oct 29, 2027 — Unit 3: Fractions, Decimals, Ratios & Averages

Fraction Sense **Core:** See fractions as numbers you can compare and combine cleverly.

Mon Oct 25
Simplest form
TRY Write $\frac{36}{48}$ in simplest form.

Tue Oct 26
Comparing fractions
TRY Which is larger: $\frac{5}{8}$ or $\frac{6}{11}$?

Wed Oct 27
Fraction of a number
TRY What is $\frac{2}{3}$ of 27?

Thu Oct 28
Clever fraction sums
TRY Compute $\frac{1}{1\times2}+\frac{1}{2\times3}+\cdots+\frac{1}{9\times10}$.

Fri Oct 29
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 12 • Nov 1–Nov 5, 2027 — Unit 3: Fractions, Decimals, Ratios & Averages

Fraction Word Problems **Core:** Bar models turn fraction stories into pictures.

Mon Nov 1
Add & subtract mixed numbers
TRY $2\frac{3}{4}+1\frac{5}{6}=?$

Tue Nov 2
Multiply fractions
TRY Compute $\frac{3}{4}\times\frac{8}{9}$.

Wed Nov 3
Fraction of the remaining
TRY Class of 36: $\frac{1}{3}$ walk to school, $\frac{1}{4}$ of the rest bike. How many bike?

Thu Nov 4
Work backward with fractions
TRY Mia spent $\frac{1}{3}$ of her money, then $\frac{1}{4}$ of the rest, leaving $24. Start?

Fri Nov 5
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 13 • Nov 8–Nov 12, 2027 — Unit 3: Fractions, Decimals, Ratios & Averages

Decimals & Percents **Core:** Move fluently among fractions, decimals and percents.

Mon Nov 8
Decimal multiplication
TRY Compute 3.6×0.25.

Tue Nov 9
Fractions ↔ decimals
TRY Write $\frac{7}{8}$ as a decimal.

Wed Nov 10
Percent of a number
TRY What is 35% of 80?

Thu Nov 11
HOLIDAY: Veterans Day
No class. Do this skill's 10 problems on the weekend:
Discounts
TRY A $60 jacket is 25% off. What is the sale price?

Fri Nov 12
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 14 • Nov 15–Nov 19, 2027 — Unit 3: Fractions, Decimals, Ratios & Averages

Ratio & Rate **Core:** Ratio tables, unit rates and distance = speed × time.

Mon Nov 15
Ratio tables
TRY Boys : girls = 3 : 5. There are 15 boys. How many girls?

Tue Nov 16
Sharing in a ratio
TRY Share 56 marbles in the ratio 3 : 4. How big is the larger share?

Wed Nov 17
Unit rate & best buy
TRY Better buy: 6 pens for $4.20 or 8 pens for $5.20?

Thu Nov 18
Speed, distance, time
TRY Cars 300 km apart drive toward each other at 70 and 80 km/h. When do they meet?

Fri Nov 19
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Thanksgiving Break • Nov 22–Nov 26, 2027 • no class (not counted in the 34 weeks)

WEEK 15 • Nov 29–Dec 3, 2027 — Unit 3: Fractions, Decimals, Ratios & Averages

Averages **Core:** Average means "level it off"; total = average × count.

Mon Nov 29
Computing the mean
TRY Find the mean of 12, 15, 9, 18, 21.

Tue Nov 30
Missing value
TRY The mean of 4 numbers is 25. Three are 20, 28, 31. Find the fourth.

Wed Dec 1
Changing averages
TRY 5 tests average 82. What 6th score makes the average 84?

Thu Dec 2
Evenly spaced lists
TRY Find the average of 21, 23, 25, . . . , 59.

Fri Dec 3
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 16 • Dec 6–Dec 10, 2027 — Unit 4: Patterns & Algebraic Thinking

Number Patterns & Sequences **Core:** Find the rule, then count and add terms.

Mon Dec 6
Find the rule
TRY Next two terms: 3, 7, 12, 18, 25, . . .

Tue Dec 7
The nth term
TRY Find the 30th term of 5, 9, 13, 17, . . .

Wed Dec 8
Counting terms (fencepost)
TRY How many numbers are in the list 7, 11, 15, . . . , 199?

Thu Dec 9
Sum of a sequence
TRY Find $7 + 11 + 15 + \cdots + 199$.

Fri Dec 10
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 17 • Dec 13–Dec 17, 2027 — Unit 4: Patterns & Algebraic Thinking

Growing Shape Patterns — **Core:** Turn a picture pattern into a table, then a rule in words.

Mon Dec 13
Toothpick patterns
TRY 1 square uses 4 toothpicks; each new square in the row adds 3. Toothpicks for 10 squares?

Tue Dec 14
Triangular & square numbers
TRY What is the 10th triangular number?

Wed Dec 15
Far figures
TRY In the same toothpick pattern, how many toothpicks for 50 squares?

Thu Dec 16
Reverse questions
TRY In the same pattern, how many squares use exactly 100 toothpicks?

Fri Dec 17
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Winter Break • Dec 20–Dec 24, 2027 • no class (not counted in the 34 weeks)

Winter Break • Dec 27–Dec 31, 2027 • no class (not counted in the 34 weeks)

WEEK 18 • Jan 3–Jan 7, 2028 — Unit 4: Patterns & Algebraic Thinking

Bar-Model Word Problems — **Core:** Draw bars first; the arithmetic follows.

Mon Jan 3
Sum & difference
TRY Two numbers add to 50 and differ by 12. Find them.

Tue Jan 4
Times as many
TRY Ann has 3 times as many stickers as Ben. Together: 48. How many does Ann have?

Wed Jan 5
Age problems
TRY Mom is 36, her son is 8. In how many years will Mom be 3 times his age?

Thu Jan 6
Before & after
TRY A has 50 marbles, B has 30. How many must A give B so they are equal?

Fri Jan 7
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 19 • Jan 10–Jan 14, 2028 — Unit 4: Patterns & Algebraic Thinking

Assumption & Work Backward — **Core:** Assume one case, then fix the difference.

Mon Jan 10
Chickens & rabbits
TRY 20 heads and 56 legs. How many rabbits?

Tue Jan 11
Two kinds of coins
TRY 15 nickels and dimes are worth $1.10. How many dimes?

Wed Jan 12
Multi-step backward
TRY I gave away half my cards, then 6 more, and have 10 left. How many at first?

Thu Jan 13
Guess-check-improve tables
TRY Two consecutive whole numbers multiply to 506. Find them.

Fri Jan 14
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 20 • Jan 17–Jan 21, 2028 — Unit 4: Patterns & Algebraic Thinking

Mid-Year Mock Contest & Review — **Core:** Review Units 1–4 and practice pacing a full contest.

Mon Jan 17
HOLIDAY: MLK Jr. Day
No class. Do this skill's 10 problems on the weekend:
Review: fractions, decimals, percents
TRY Order from least to greatest: 0.6, $\frac{5}{8}$, 62%.

Tue Jan 18
Review: rate & averages
TRY A car goes 180 km in 2.5 hours. What is its speed?

Wed Jan 19
Review: patterns & bar models
TRY Two numbers add to 84 and differ by 16. Find them.

Thu Jan 20
Pacing: skip & return
TRY 15-minute set. Warm-up: average of 18, 24, 30?

Fri Jan 21
Mid-Year Mock Contest
Full-length, Units 1–4. Then sort errors by topic in the error log.

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WEEK 21 • Jan 24–Jan 28, 2028 — Unit 5: Geometry & Spatial Reasoning

Angles & Polygons **Core:** Angle facts on lines, in triangles and in polygons.

Mon Jan 24
Angle types & turns
TRY How many degrees does the minute hand turn in 20 minutes?

Tue Jan 25
Angles on a line
TRY Two angles form a straight line. One is $47°$. Find the other.

Wed Jan 26
Triangle angle sum
TRY A triangle has angles $48°$ and $75°$. Find the third.

Thu Jan 27
Polygon angle sums
TRY What is the sum of the interior angles of a hexagon?

Fri Jan 28
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 22 • Jan 31–Feb 4, 2028 — Unit 5: Geometry & Spatial Reasoning

Perimeter & Area **Core:** Rectangles, triangles, and figures built from them.

Mon Jan 31
Rectangles
TRY A rectangle has perimeter 30 cm and length 9 cm. Find its area.

Tue Feb 1
Triangles & parallelograms
TRY A triangle has base 12 cm and height 7 cm. Find its area.

Wed Feb 2
Composite figures
TRY A 10×8 rectangle has a 4×3 corner cut off. Area left?

Thu Feb 3
Same area, different perimeter
TRY How many rectangles with whole-number sides have area 24?

Fri Feb 4
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 23 • Feb 7–Feb 11, 2028 — Unit 5: Geometry & Spatial Reasoning

Area by Cut & Rearrange **Core:** Shaded area = whole − unshaded; move pieces to see it.

Mon Feb 7
Subtract areas
TRY A 4×6 rectangle is cut from a 10×10 square. Area left?

Tue Feb 8
Half-of-rectangle triangles
TRY A triangle uses one 8-unit side of an 8×5 rectangle and a point on the opposite side. Area?

Wed Feb 9
Grid areas
TRY Triangle with corners (0,0), (6,0), (2,4) on grid paper. Area?

Thu Feb 10
Overlapping figures
TRY Two 6×6 squares overlap in a 2×6 strip. Total area covered?

Fri Feb 11
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 24 • Feb 14–Feb 18, 2028 — Unit 5: Geometry & Spatial Reasoning

Counting Figures **Core:** Count by size or by corners so nothing is missed.

Mon Feb 14
Segments
TRY 6 points lie on a line. How many segments do they make?

Tue Feb 15
Squares in a grid
TRY How many squares of all sizes are in a 4×4 grid?

Wed Feb 16
Rectangles in a grid
TRY How many rectangles are in a 2×3 grid?

Thu Feb 17
Triangles in a figure
TRY From the top corner of a triangle, draw 3 lines to the base. How many triangles?

Fri Feb 18
Weekly Quiz
10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 25 • Feb 21–Feb 25, 2028 — Unit 5: Geometry & Spatial Reasoning

Cubes, Volume & Surface Area — **Core:** Count cubes for volume and faces for surface area.

Mon Feb 21

HOLIDAY: Presidents' Day

No class. Do this skill's 10 problems on the weekend:

Volume

TRY Find the volume of a $5 \times 4 \times 3$ cm box.

Tue Feb 22

Surface area

TRY Find the surface area of the same box.

Wed Feb 23

Stacked-cube solids

TRY Remove one corner cube from a $2 \times 2 \times 2$ cube. New surface area?

Thu Feb 24

Painted cube

TRY A $3 \times 3 \times 3$ cube is painted, then cut into 27 cubes. How many have exactly 2 painted faces?

Fri Feb 25

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 26 • Feb 28–Mar 3, 2028 — Unit 5: Geometry & Spatial Reasoning

Nets & Views — **Core:** Fold, unfold and look from different sides.

Mon Feb 28

Cube nets

TRY How many different nets does a cube have?

Tue Feb 29

Opposite faces

TRY Opposite faces of a die add to 7. Which number is opposite 2?

Wed Mar 1

Front, top, side views

TRY 3 cubes in a row with 1 cube on the middle one. How many squares in the front view?

Thu Mar 2

Building from views

TRY Front view and side view are both 2×2 squares. Fewest cubes possible?

Fri Mar 3

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 27 • Mar 6–Mar 10, 2028 — Unit 6: Counting, Probability & Logic

Systematic Counting — **Core:** Organize the count: lists, trees, and endpoints.

Mon Mar 6

Lists & tree diagrams

TRY How many ways can A, B, C be arranged in a row?

Tue Mar 7

Avoid double counting

TRY 8 people each shake hands once with everyone else. How many handshakes?

Wed Mar 8

Counting a range

TRY How many whole numbers are from 15 to 87, inclusive?

Thu Mar 9

Page-number digits

TRY How many digits are used to number pages 1 to 120?

Fri Mar 10

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 28 • Mar 13–Mar 17, 2028 — Unit 6: Counting, Probability & Logic

Multiplication Principle — **Core:** Choices in stages multiply.

Mon Mar 13

Outfits & menus

TRY 4 shirts and 3 pants. How many outfits?

Tue Mar 14

Codes

TRY 3-digit codes from 0–9: how many if digits may repeat? If not?

Wed Mar 15

Arrangements in a line

TRY In how many ways can 5 kids stand in a line?

Thu Mar 16

Restrictions

TRY How many 3-digit numbers have all different digits?

Fri Mar 17

Weekly Quiz

10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

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WEEK 29 • Mar 20–Mar 24, 2028 — Unit 6: Counting, Probability & Logic

Venn Diagrams **Core:** Count overlaps once: add, subtract, add back.

Mon Mar 20	Tue Mar 21	Wed Mar 22	Thu Mar 23	Fri Mar 24
Two-set Venn diagrams	**Both & neither**	**Three-set Venn diagrams**	**Divisible by 2 or 3**	**Weekly Quiz**
TRY 30 students: 18 like cats, 15 dogs, 7 both. How many like only cats?	TRY Same class: how many like neither?	TRY 40 kids: 20 soccer, 18 swim, 15 chess; 7 S&Sw, 5 S&C, 4 Sw&C, 2 all. How many do none?	TRY How many numbers from 1 to 100 are divisible by 2 or 3?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

Spring Break • Mar 27–Mar 31, 2028 • no class (not counted in the 34 weeks)

WEEK 30 • Apr 3–Apr 7, 2028 — Unit 6: Counting, Probability & Logic

Probability **Core:** Probability = favorable outcomes ÷ all equally likely outcomes.

Mon Apr 3	Tue Apr 4	Wed Apr 5	Thu Apr 6	Fri Apr 7
Outcomes as fractions	**Spinners**	**Two dice**	**At least one**	**Weekly Quiz**
TRY A bag has 3 red, 5 blue, 2 green. P(blue)?	TRY A spinner has 8 equal parts numbered 1–8. P(multiple of 3)?	TRY Roll two dice. P(sum is 7)?	TRY Flip 3 coins. P(at least one head)?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 31 • Apr 10–Apr 14, 2028 — Unit 6: Counting, Probability & Logic

Logic & Pigeonhole **Core:** Reason from clues; guarantee results with pigeonholes.

Mon Apr 10	Tue Apr 11	Wed Apr 12	Thu Apr 13	Fri Apr 14
Logic grids	**Truth-tellers & liars**	**Pigeonhole principle**	**Weighing puzzles**	**Weekly Quiz**
TRY Amy, Ben, Cal own a cat, dog, fish. Amy is allergic to fur; Ben's pet barks. Who has the cat?	TRY A: "B lies." B: "C lies." C: "A and B both lie." Who tells the truth?	TRY Red, blue and black socks in the dark. How many to be sure of a matching pair?	TRY 8 coins, one lighter. Fewest balance weighings to be sure to find it?	10 mixed problems: 8 on this week's skills + 2 review. Correct and log every miss.

WEEK 32 • Apr 17–Apr 21, 2028 — Contest Season Wrap-Up

Contest Strategy & AMC 8 Preview **Core:** Estimate, eliminate, and check with a second method.

Mon Apr 17	Tue Apr 18	Wed Apr 19	Thu Apr 20	Fri Apr 21
Estimate & eliminate	**Check a second way**	**AMC 8 style (Problems 1–10)**	**Timed set: 10 in 20 min**	**Weekly Quiz**
TRY 49×21 is closest to: 800, 1,000, or 1,200?	TRY Find $\frac{3}{4}$ of 96 two different ways.	TRY Compute $(2 + 4 + 6) - (1 + 3 + 5)$.	TRY How many multiples of 7 are between 100 and 200?	10 mixed contest problems, timed (25 min).

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WEEK 33 • Apr 24–Apr 28, 2028 — Contest Season Wrap-Up

Full Mock Contest & Review — **Core:** Simulate contest day and learn from every error.

Mon Apr 24
Review: number theory & counting
TRY What is the units digit of 2^{2027}?

Tue Apr 25
Review: geometry
TRY What is the sum of the interior angles of a pentagon?

Wed Apr 26
Review: fractions, ratios, patterns
TRY Share 90 in the ratio 2 : 3 : 4.

Thu Apr 27
Full Mock Contest
Full-length mixed contest under real time limits.

Fri Apr 28
Error-log review
Redo every missed problem from the mock; write the key idea for each.

WEEK 34 • May 1–May 5, 2028 — Contest Season Wrap-Up

Final Contest & Celebration — **Core:** Show what you have learned and set next year's goals.

Mon May 1
Light review: favorite strategies
TRY Find $1 + 3 + 5 + \cdots + 19$.

Tue May 2
Mixed warm-up set
TRY Find the GCF of 84 and 126.

Wed May 3
Final Contest
Full-length contest covering the whole year.

Thu May 4
Contest review
Go over solutions together; students present one problem each.

Fri May 5
Celebration & reflection
Certificates, progress review, and goals for next year.

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ANSWER KEY

Answers to the Daily "Try" Problems

Week 1
Mon: $13
Tue: 35 m
Wed: 3 ways
Thu: 12

Week 2
Mon: 70,000
Tue: 8,530
Wed: 9
Thu: 447 + 256 = 703

Week 3
Mon: 765
Tue: 3,700
Wed: 4,554
Thu: 1,275

Week 4
Mon: 10
Tue: $4 + 4 - 4 \div 4$
Wed: 5
Thu: 13

Week 5
Mon: Yes; yes
Tue: Yes (digit sum 18)
Wed: Yes $(5 - 0 + 7 - 1 = 11)$
Thu: 5

Week 6
Mon: 9
Tue: 5
Wed: $2^3 \times 3^2 \times 5$
Thu: 108

Week 7
Mon: 24
Tue: 36
Wed: 72
Thu: 9:24

Week 8
Mon: 12
Tue: 14
Wed: 4, 9, 25, 49
Thu: 10

Week 9
Mon: 4
Tue: D
Wed: Wednesday
Thu: 9

Week 10
Mon: 4,000
Tue: 3 or 9
Wed: 3
Thu: 120

Week 11
Mon: $\frac{3}{4}$
Tue: $\frac{5}{8}$
Wed: 18
Thu: $\frac{9}{10}$

Week 12
Mon: $4\frac{7}{12}$
Tue: $\frac{2}{3}$
Wed: 6
Thu: $48

Week 13
Mon: 0.9
Tue: 0.875
Wed: 28
Thu: $45

Week 14
Mon: 25
Tue: 32
Wed: 8 for $5.20
Thu: 2 hours

Week 15
Mon: 15
Tue: 21
Wed: 94
Thu: 40

Week 16
Mon: 33, 42
Tue: 121
Wed: 49
Thu: 5,047

Week 17
Mon: 31
Tue: 55
Wed: 151
Thu: 33

Week 18
Mon: 31 and 19
Tue: 36
Wed: 6 years
Thu: 10

Week 19
Mon: 8
Tue: 7
Wed: 32
Thu: 22 and 23

Week 20
Mon: 0.6, 62%, $\frac{5}{8}$
Tue: 72 km/h
Wed: 50 and 34
Thu: 24

Week 21
Mon: 120°
Tue: 133°
Wed: 57°
Thu: 720°

Week 22
Mon: 54 cm^2
Tue: 42 cm^2
Wed: 68
Thu: 4

Week 23
Mon: 76
Tue: 20
Wed: 12
Thu: 60

Week 24
Mon: 15
Tue: 30
Wed: 18
Thu: 10

Week 25
Mon: 60 cm^3
Tue: 94 cm^2
Wed: 24 (unchanged)
Thu: 12

Week 26
Mon: 11
Tue: 5
Wed: 4
Thu: 4

Week 27
Mon: 6
Tue: 28
Wed: 73
Thu: 252

Week 28
Mon: 12
Tue: 1,000; 720
Wed: 120
Thu: 648

Week 29
Mon: 11
Tue: 4
Wed: 1
Thu: 67

Week 30
Mon: $\frac{1}{2}$
Tue: $\frac{1}{4}$
Wed: $\frac{1}{6}$
Thu: $\frac{7}{8}$

Week 31
Mon: Cal
Tue: Only B
Wed: 4
Thu: 2

Week 32
Mon: 1,000
Tue: 72
Wed: 3
Thu: 14

Week 33
Mon: 8
Tue: 540°
Wed: 20, 30, 40

Week 34
Mon: 100
Tue: 42

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